

KUAN-HSUN TU

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EDUCATION

National Taiwan University

Sep. 2021 – Present

Bachelor of Science in Computer Science and Information Engineering

Taipei, Taiwan

- **Expected Graduation:** June 2026
- **Research Focus:** Embodied AI, Vision-Language-Action models, and large-scale learning systems

RESEARCH EXPERIENCE

Embodied AI Lab

Sep. 2024 – Present

Undergraduate Researcher – Advisor: Dr. Tsung-Wei Ke

Taipei, Taiwan

- Conducting research on VLM-guided exploration, video-to-digital-twin policy learning, and flow-based policy improvement.
- Contributed to a joint project with Carnegie Mellon University on video-to-digital-twin benchmarking (ICLR 2026).

PUBLICATIONS

- [1] J. Hsieh, **K.-H. Tu**, K.-H. Hung and T.-W. Ke, “DexMan: Learning Bimanual Dexterous Manipulation from Human and Generated Videos,” *arXiv:2510.08475*.
- [2] Y. Jangir, Y. Zhang, K. Yamazaki, C. Zhang, **K.-H. Tu**, T.-W. Ke, L. Ke, Y. Bisk, and K. Fragkiadaki, “RobotArena ∞ : Unlimited Robot Benchmarking via Real-to-Sim Translation,” *arXiv:2510.23571*.
- [3] **K.-H. Tu***, H.-C. Liu*, P. Y. Wu*, S.-H. Sun, and T.-W. Ke, “Split Decisions: VLM-Guided Action Sampling for Efficient RL Exploration,” *under review at ICLR 2026*.
- [4] C.-S. Chiang*, **K.-H. Tu***, J.-W. Liao*, and T.-W. Ke, “COT-FM: Cluster-wise Optimal Transport Flow Matching,” *under review at CVPR 2026*.

AWARDS & HONORS

SC25 Student Cluster Competition – Overall Winner

Nov. 2025

ACM/IEEE Supercomputing Conference

St. Louis, USA

- Achieved **561.8 TFLOPs** on HPL benchmark using two GPU nodes with 16 NVIDIA H200 NVLs
- Served as the cluster system administrator, contributing to HPC benchmarking (HPL, HPL-MxP), Reproducibility Challenge, and Mystery Application tracks

APAC HPC-AI Competition – 2nd Place & Best AI Performance

Nov. 2025

Asia Pacific Advanced Network

Singapore

- Secured 2nd place overall and received Best AI Performance award
- Optimized HPC cluster for AI workloads and demonstrated superior AI benchmark performance

TECHNICAL EXPERIENCE

Lead System Administrator

July 2025 – Present

System Administration Team, Department of CSIE, NTU

Taipei, Taiwan

- Appointed to lead department-wide infrastructure governance and continuity planning
- Supervised junior system administrators and optimized workflows across compute and services supporting 1,000+ faculty, staff, and students

HPC Student Cluster Engineer

July 2025 – Nov. 2025

NTU Student Cluster Competition Team – Collaborating with Dr. Chun-Yi Lee

Taipei, Taiwan

- Built and optimized a full HPC cluster stack from scratch, benchmarked with MLPerf and HPL, achieving SC25 competition-level performance

TEACHING EXPERIENCE

Lead Teaching Assistant

Sep. 2025 – Present

Network Administration & System Administration, Department of CSIE, NTU

Taipei, Taiwan

- Coordinated a TA team in grading, lab design, and assignment workflows for 80+ students
- Supervised other TAs and ensured consistency in lab delivery and exam evaluation

Teaching Assistant

Sep. 2025 – Present

Parallel Programming, Department of CSIE, NTU

Taipei, Taiwan

- Set up and maintained lab machines, including system configuration and software environments
- Provided guidance to students on debugging, cluster usage, and automation workflows